

ANKARA BOLGE, Turkey

WMO: 171300

Lat: 39.973N

Lon: 32.863E

Elev: 891

StdP: 91.07

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.1	-5.8	-13.0	1.4	-3.6	-10.7	1.7	-2.0	6.4	4.6	5.7	4.4	1.9	50	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.0	34.8	17.0	33.1	16.8	31.7	16.5	19.1	29.4	18.4	28.8	17.8	28.1	2.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.7	22.4	15.0	11.9	21.8	14.1	11.2	21.3	58.3	29.3	55.8	28.7	53.7	28.0	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	4.9	DB	-11.8	37.6	2.9	1.6	-13.8	38.7	-15.5	39.7	-17.1	40.6	-19.2	41.8	(4)
			WB	-12.4	20.5	2.7	1.2	-14.3	21.4	-15.9	22.1	-17.4	22.7	-19.4	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	2.2	3.9	7.4	12.5	16.9	21.0	24.4	24.8	20.2	14.2	8.6	3.5
	DBStd	8.65	4.18	4.48	3.98	3.64	3.23	2.79	2.56	2.54	3.70	3.66	3.59	4.02
	HDD10.0	811	243	174	98	14	0	0	0	0	0	8	69	203
	HDD18.3	2407	502	405	339	179	67	7	0	0	23	132	293	459
	CDD10.0	2033	1	3	17	89	215	331	445	459	306	140	26	2
	CDD18.3	587	0	0	0	3	23	88	187	201	80	6	0	0
	CDH23.3	6426	0	0	1	52	254	887	2109	2208	846	69	0	0
	CDH26.7	2608	0	0	0	4	47	293	941	1008	307	8	0	0
Wind	WSAvg	2.1	1.8	2.0	2.0	2.1	2.1	2.3	2.6	2.8	2.2	1.7	1.5	1.7
Precipitation	PrecAvg	411	43	36	43	46	53	35	13	13	18	31	33	46
	PrecMax	620	102	86	96	111	122	122	67	81	78	173	113	117
	PrecMin	242	5	2	9	4	2	0	0	0	0	1	1	1
	PrecStd	83	25	22	22	26	27	25	13	17	21	27	22	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	17.7	21.6	27.2	29.9	33.4	37.5	37.0	34.2	27.6	21.3	15.5
		MCWB	8.1	9.2	10.1	13.0	14.5	17.0	17.3	17.0	15.5	13.0	11.3	9.2
	2%	DB	10.7	14.7	18.6	24.9	27.9	31.6	34.8	35.0	32.0	25.2	19.6	12.3
		MCWB	6.5	8.0	9.1	12.0	14.5	16.5	17.2	17.1	15.6	13.3	11.0	8.1
	5%	DB	9.1	12.3	16.6	22.5	26.3	29.9	33.0	33.4	30.3	23.3	17.2	10.8
		MCWB	5.8	7.1	8.3	11.5	14.1	16.0	16.9	17.0	15.2	12.7	9.9	7.4
	10%	DB	7.7	10.5	14.5	20.2	24.3	28.2	31.4	31.7	28.2	21.2	14.9	9.1
		MCWB	5.2	6.3	7.7	10.6	13.7	15.9	16.8	16.8	14.6	12.3	8.9	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	10.5	11.1	13.9	16.6	18.7	20.2	20.5	17.9	15.6	13.5	10.2
		MCDB	12.4	16.6	18.2	24.3	24.8	28.8	30.6	29.5	28.3	19.5	18.7	13.6
	2%	WB	7.1	8.6	9.9	13.0	15.7	17.8	19.2	19.3	16.8	14.6	11.6	8.9
		MCDB	9.9	12.9	16.9	22.8	24.9	28.0	29.4	29.5	26.7	20.4	18.0	11.5
	5%	WB	6.2	7.6	9.0	12.2	15.0	17.2	18.4	18.6	16.2	13.9	10.2	7.8
		MCDB	8.5	11.4	14.9	20.7	24.0	26.8	28.7	29.0	26.7	20.6	15.7	10.2
	10%	WB	5.4	6.6	8.1	11.3	14.3	16.6	17.8	18.0	15.6	13.3	9.4	6.8
		MCDB	7.4	10.1	13.4	18.7	22.4	25.7	28.2	28.2	25.9	19.5	14.0	9.0
Mean Daily Temperature Range	5% DB	MDBR	6.0	8.2	9.7	11.3	11.5	12.4	13.1	13.0	12.7	11.2	10.4	6.9
		MCDBR	8.0	11.4	13.5	14.9	14.1	14.3	14.6	14.4	15.0	14.2	13.3	9.3
		MCWBR	4.9	6.0	6.4	5.8	5.1	4.3	4.1	4.2	5.0	5.9	6.5	5.7
	5% WB	MCDBR	7.1	10.0	11.2	13.1	12.7	12.9	13.3	13.0	13.0	10.8	11.6	8.1
		MCWBR	4.5	5.4	5.6	5.4	5.0	4.4	4.3	4.2	4.5	4.7	5.9	5.0
Clear-Sky Solar Irradiance	taub	0.304	0.330	0.373	0.409	0.389	0.369	0.367	0.365	0.345	0.346	0.315	0.305	
	taud	2.383	2.332	2.234	2.159	2.284	2.360	2.364	2.347	2.402	2.397	2.454	2.442	
	Ebn at Noon	858	882	876	864	890	905	904	897	896	854	841	830	
	Edh at Noon	84	103	126	146	132	122	121	119	106	95	78	74	
All-Sky Solar Radiation	RadAvg	1.87	2.88	4.03	5.16	6.14	6.99	7.42	6.61	5.20	3.55	2.45	1.66	
	RadStd	0.23	0.42	0.36	0.47	0.45	0.58	0.31	0.27	0.35	0.26	0.24	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page